2024 Spr. 高代下 midterm (H).

Problem 1. Let k be a field and $\mathrm{Fin\,Vect}_k$ the collection of all finite dimensional vector spaces over k. Let $S := \mathrm{Fin\,Vect}_k$. Consider a relation $\sim$ over S defined by

$$(V,W)\sim(V',W') \text{ if and only if } V\oplus W'\cong V'\oplus W$$

(i) Prove that $\sim$ is an equivalent relation. In other words, (1) $a\sim a$; (2) $a\sim b$ implies $b\sim a$; (3) $a\sim b$ and $b\sim c$ implies $a\sim c$.

(ii) Calculate $S\times S/\sim$. In other words, find all equivalence classes, and find a representative (will be refered to as distinguished represantative in other subproblems) for each equivalence class.

(iii) For $(V,W),(V',W')\in S\times S$, let

$$(V,W)\oplus(V',W') := (V\oplus V', W\oplus W')$$

Prove that it induces a well-defined map on the quotient $S\times S/\sim$. Moreover, describe the map $\oplus$ explicitly using the representatives you found in (ii) (i.e. for each distinguished representative (V,W) and each distinguished representative (V',W'), find a distinguished representative (V'',W'') such that $(V,W)\oplus(V',W')$ is equivalent to (V'',W'')).

(iv) For $(V,W),(V',W')\in S\times S$, let

$$(V,W)\otimes(V',W') := ((V\otimes V')\oplus(W\otimes W'),(V\otimes W')\oplus(V'\otimes W)).$$

Prove that it induces a well-defined map on the quotient $S\times S/\sim$. Moreover, describe the map $\otimes$ explicitly using the representatives you found in (ii).

Problem 2. Let A, B be submodules of some k-module X. Prove that

$$\frac{A+B}{B}\cong\frac{A}{A\cap B}$$

Problem 3. Let k be a field and consider $k[x]$. Let $f\in k[x]$ be a fixed monic polynomial. Recall that, for $g\in k[x]$ such that f and g are coprime in $k[x]$, if we run Euclidean algorithm on f and g, we can find $p,q\in k[x]$ such that

$$pf+qg=1$$

Now, consider $R := k[x]/(f)$. Then, the formula above gives

$$qg=1$$

in R, where $q, g, 1$ are treated as their equivalence classes in $R=k[x]/(f)$. In ~~order~~ other words, for every g that is coprime with f in $k[x]$, there exists g^{-1} in R. Moreover, this g^{-1} can be calculated using Euclidean algorithm. Let $f=x^4+x^3+x^2+x+1$. For each of the following g, (1) determine if f and g are coprime in $k[x]$, (2) calculate g^{-1} if they are coprime in R.

(i) $g=1$ (ii) $g=x$ (iii) $g=x^2$ (iv) $g=x^3$.

Problem 4. Let k be a field, and let $a_0, \cdots, a_{n-1} \in k$. Consider the $n \times n$ matrix A given by

$$A_{ij} = \begin{cases} a_{j-i}, & i \leq j, \\ 0, & i > j. \end{cases}$$

(i) Calculate the minimal polynomial of A when $a_1 = 1$ (assume that $n \geq 2$).

(ii) Calculate the minimal polynomial of A when $a_1 = 0$ and $a_2 = 1$ (assume that $n \geq 3$).

(iii) Give a conjectural formula for the minimal polynomial of A.

(iv) Prove the conjecture.

Problem 5. For a k-module M, let $\iota_M : M \to M^{**}$ given by $m \mapsto ev_m$, where $ev_m : M^* \to k$ given by $f \mapsto f(m)$. For $f \in \mathrm{Hom}(M, N)$, let $f^{**} \in \mathrm{Hom}(M^{**}, N^{**})$ be the double dual of f.

(i) Prove that $\iota_N \circ f = f^{**} \circ \iota_M$ as elements in $\mathrm{Hom}(M, N^{**})$.

(ii) Suppose that M has a basis $b_1, \cdots, b_d$. Let $b_1^*, \cdots, b_d^* \in M^*$ be the dual basis of $b_1, \cdots, b_d$. Let $b_1^{**}, \cdots, b_d^{**} \in M^{**}$ be the dual basis of $b_1^*, \cdots, b_d^*$. Prove that $\iota_M(b_i) = b_i^{**}$.

Problem 6. Let M be a finite free k-module equipped with a homomorphism $\Delta : M \to M \otimes M$. Let $id_M \in \mathrm{Hom}(M, M)$ be the identity map on M. The pair (M, Δ) is called cocommutative if the following diagram commutes.

$$\begin{array}{ccc} M & \xrightarrow{\Delta} & M \otimes N \\ \downarrow \Delta & & \downarrow \Delta \otimes id_M \\ M \otimes M & \xrightarrow{id_M \otimes \Delta} & M \otimes M \otimes M. \end{array}$$

(Recall that for $f \in \mathrm{Hom}(A, A')$ and $g \in \mathrm{Hom}(B, B')$, we define $f \otimes g \in \mathrm{Hom}(A \otimes B, B' \otimes B')$ given by $(f \otimes g)(a \otimes b) := f(a) \otimes g(b)$.)

(i) Suppose that M has basis a, b, c, d, and define the homomorphism Δ as follows.

$$\Delta : M \to M \otimes M$$
$$a \mapsto a \otimes a + b \otimes c$$
$$b \mapsto a \otimes b + b \otimes d$$
$$c \mapsto c \otimes a + d \otimes c$$
$$d \mapsto c \otimes b + d \otimes d$$

Prove that it is cocommutative.

(ii) Consider the map $\phi \in \mathrm{Hom}(M^* \otimes M^*, M^*)$ given by

$$\phi(f \otimes g) = (m \mapsto (f \otimes g)(\Delta(m))).$$

We call the pair (M^*, ϕ) commutative if $\phi(f \otimes g) = \phi(g \otimes f)$ for all $f, g \in M^*$. Prove that, (M^*, ϕ) is commutative if and only if (M, Δ) is cocommutative.

Problem 7. Let k be an integral domain (i.e. for $a,b\in k$, $ab=0$ implies either $a=0$ or $b=0$). Let M be a k-module, and $B\in(M,M^*)$. We call B

- reflexive if $B(x,y)=0$ implies that $B(y,x)=0$ for all $x,y\in M$.
- symmetric if $B(x,y)=B(y,x)$ for all $x,y\in M$.
- alternating if $B(x,y)=0$ for all $x\in M$.

The goal of this problem is to prove that:

B is reflexive if and only if B is either symmetric or alternating.

The sufficiency is easy, and you do not need to provide a proof. You are asked to prove the necessity by proving the following statements. Assume now that B is reflexive.

(i) For all $x,y,z\in M$,

$$B(x,y)B(z,x)=B(x,z)B(y,x)$$

(ii) For all $x,y\in M$, if $B(x,x)\neq 0$, then

$$B(x,y)=B(y,x)$$

(iii) Suppose that for some $x,y\in M$, $B(x,y)\neq B(y,x)$. If $B(z,z)\neq 0$ for some $z\in M$, then

$$B(x+z,x+z)=0$$

(iv) Prove that if B is either symmetric or alternating.

Problem 8. Let M be a bilinear space, where $M=k^n$ as a k-module (k is a field where 2 is invertible). Suppose that

$$\langle e_i,e_j\rangle=j-i.$$

for $1\leq i\leq j\leq n$.

(i) Assume that $n=5$ and M is symmetric. Find an explicit basis for M under which the Gram matrix is diagonal.

(ii) Assume that $n=5$ and M is alternating. Find an explicit basis for M under which the Gram matrix is block diagonal matrix, where each block is either the 2×2 alternating matrix with diagonal entries 0 and anti-diagonal entries ± 1, or 1×1 matrix with a single entry 0.